

* Error which I made: I wrote

$$C_{j+1} = \gamma \frac{2(1+j+l) - 2}{(j+1)(j+2l+2)} C_j$$

But it should be

$$C_{j+1} = \frac{2\gamma(1+j+l) - 2}{(j+1)(j+2l+2)} C_j$$

$$\gamma = \frac{1}{n}$$

Okay, so to find $R_{n,l}$ we use

$$R_{n,l} = \left(\frac{r}{a_0}\right)^l e^{-\frac{r}{na_0}} \sum_{i=0}^{n-l} C_i \left(\frac{r}{a_0}\right)^i$$

- Start with $n=1$: Since $l_{\max} = n-1$
 $l=0$ is the only case

$$R_{n=1,0} = \left(\frac{r}{a_0}\right)^0 e^{-r/a_0} C_0$$

- $n=2$: means $l=0, 1$

$$R_{2,l} = \left(\frac{r}{a_0}\right)^l e^{-r/2a_0} \left[C_0 + C_1 \left(\frac{r}{a_0}\right)^1 \right]$$

(a) $l=0$:

$$R_{2,0} = e^{-r/2a_0} \left[C_0 + C_1 \left(\frac{r}{a_0}\right) \right]$$

$$C_{j+1} = \frac{\left(\frac{2}{n}(j+l+1) - 2\right)}{(j+1)(j+2l+2)} C_j$$

for C_1 $j=0$ and $n=2$

$$C_1 = \frac{\left(\frac{2}{2}(0+0+1) - 2\right)}{2(0+1)} = -\frac{1}{2} C_0$$

thus

$$R_{2,0} = C_0 e^{-r/2a_0} \left(1 - \frac{1}{2} \left(\frac{r}{a_0}\right) \right)$$

(b) $l=1$

$$R_{2,1} = \left(\frac{r}{a_0}\right)^2 e^{-r/2a_0} \left(C_0 + C_1 \left(\frac{r}{a_0}\right)\right)$$

$$C_1 = \frac{\left(\frac{2}{2}(0+1+1)-2\right)}{2(1+1)} C_0$$

Note: Each value of l has a different C_l value

$$= \frac{(2-2)}{4} C_0$$

$$= 0$$

$$R_{2,1} = C_0 \left(\frac{r}{a_0}\right)^2 e^{-r/2a_0}$$

For $n=3$: $l=0, 1, 2$

$$R_{3,2} = \left(\frac{r}{a_0}\right)^2 e^{-r/3a_0} \left(C_0 + C_1 \left(\frac{r}{a_0}\right) + C_2 \left(\frac{r}{a_0}\right)^2\right)$$

$$C_1 = \frac{\left(\frac{2}{3}(0+2+1)-2\right)}{2(2+1)} C_0$$

$$= \left(\frac{2-2}{6}\right) C_0 = 0$$

$C_1 = 0$ and so C_2 is too since

$$C_2 = () C_1$$

$$C_1 = 0$$

$$C_2 = 0$$

$$R_{2,1} = C_0 \left(\frac{r}{a_0}\right)^2 e^{-r/3a_0}$$

To find the constant C_0

$$1 = \int_0^\infty dr r^2 |C_0|^2 \left(\frac{r}{a_0}\right)^4 e^{-\frac{2}{3}\frac{r}{a_0}}$$

$$|C_0|^2 \int_0^\infty dr r^6 e^{-\frac{2}{3}\frac{r}{a_0}} = 1$$

a_0^u $\int$
solve this
integral
to get C_0

